OPEN AGENT HACKATHON 2026

FastTrack Termin

A faster route to a specialist appointment.
With evidence for every reason.

Track 02: “The Agent That Can Explain Why”
NVIDIA Nemotron prize submission

Berlin · Statutory-insurance patients · Prototype
Seed practices and waiting times are SAMPLE data. Not medical advice. Emergency: 112.

Andreas Krüger · Public repository: github.com/AndiKrue/fasttrack-termin · AGPL-3.0

PROBLEM AND TARGET USERS

Waiting is only part of the problem

Berlin statutory-insurance patients facing weeks of waiting may not know which route to try next.

Appointment service

What does the service offer, and which referral conditions apply?

Open consultation

Could an appointment-free consultation be an option?

GP referral

Could the GP help arrange the specialist appointment?

THE SOLUTION IN USE

Use it: web, Telegram, mini app

Web

`127.0.0.1:8765`

Watch the live step timeline. Compare routes, expected days and reasons.

Open evidence chips for SQL, rows and web source links.

Telegram

Scan the bot QR code: @FastTrackTerminBot

Ask in chat. Watch one status message update live.

Get routes with phone, map and source links.

Mini app

Open the bot’s menu button.

The same live flow inside Telegram, with theme colours and the main button.

Call or route when cited data has a phone or address.

```
cp .env.example .env    # fill keys locally; never commit .env
docker compose up --build -d
```

Compose starts app + Cloudflare HTTPS tunnel + bot. This prototype does not book appointments.

AGENT WORKFLOW AND ARCHITECTURE

The explanation has a data trail

1 · Plan →

Find the factual questions.

Nemotron Lightning 30B

2 · SQL →

Query three Zetaris sources.

Rules · practices · waits

3 · Recall ↓

Retrieve earlier findings.

Meterless H-MEM

4 · Answer ←

Rank routes; cite evidence IDs.

Nemotron Ultra 550B

5 · Verify ←

Drop unsupported routes.

Nemotron Lightning 30B

6 · Remember

Store only retained routes.

Run-linked memory

Optional Firecrawl search + Lightning extraction precedes Recall. Live steps reach web and Telegram; SQL/web rows share evidence IDs.

WHAT IS UNIQUE

A reason points to an inspectable source

Claim

A route and its factual explanation.

No invented appointment promise.

Evidence ID

`E1`

Illustrative ID

Evidence object

`SQL + row` or `URL + row`

Inspect the exact query result or a web fact and its source URL.

The verifier is instructed to keep a route only when every factual statement is supported by its cited rows.

Unknown IDs are removed; routes need a valid citation and a keep verdict. This is an LLM check, not a correctness guarantee.

ZETARIS

Three sources. One SQL surface.

Source	What the agent retrieves
`FT_RULES.rules`	Route rules, applicability and legal references
`FT_PRACTICES.practices`	Practice directory and consultation attributes
`FT_WAITS.waits`	Channel-specific waiting-time observations

Zetaris Lightning: Docker Compose platform; queries over JDBC. `zetaris/register.py` registers the seed CSV tables.

Local fallback: SQLite with the same schemas. Seed practices/waits are SAMPLE. Optional Firecrawl adds real web practice facts with URLs; the answer prefers these rows.

METERLESS H-MEM

Memory keeps a link to its origin

Recall before answering

A TypeScript bridge queries H-MEM for earlier findings relevant to the question.

Recalled findings provide context; the answer still needs current evidence rows.

Remember after verification

Retained route and reason become factual memory with a run tag and `fasttrack:<run_id>` source.

H-MEM persists state and a ledger.

Current provenance boundary: H-MEM stores text and the run link. Full SQL/rows are in the returned evidence; the JSONL fallback stores the full record.

TECHNICAL STACK AND NVIDIA NEMOTRON

Four roles, with separate model choices

Role	Configured model	Responsibility
Planner	Lightning 30B	Select factual questions
SQL	Lightning 30B	SELECTs and web search query
Verifier	Lightning 30B	Route checks; web fact extraction
Answer	Ultra 550B	Rank routes and explain why

Image: Python 3.12 + JRE 17 for Zetaris JDBC + Node for H-MEM.
Compose: app, Cloudflare quick tunnel (public HTTPS), Telegram bot.
Zetaris: `host.docker.internal:10000`; empty `ZETARIS_JDBC_JAR` uses local SQLite.

NVIDIA API · Optional Firecrawl: cloud or self-hosted · No measured gains claimed.

PRODUCT VISUALS

Show the route, then inspect the reason

Ask and compare

SCREENSHOT PLACEHOLDER A

Question + ranked route cards

Web: timeline, route, Call/Route.
Keep the SAMPLE label visible.

Telegram + mini app

SCREENSHOT PLACEHOLDER B

Live bot status + mini app

Insert actual Telegram captures.
Show evidence / source links.

Optional “Use my location” shows distance via Nominatim. Placeholders only; no live results shown.

FUTURE WORK

From sample routes to real access

Real practice data

Expand web coverage; replace SAMPLE waits with maintained data.

Track freshness and source provenance.

Booking integration

Connect supported routes to booking services.

Confirm availability and obtain patient approval before booking.

Other cities

Extend directories and validate local route information.

Keep the same evidence and verifier contract.

Next validation: live integrations, verifier failure cases, and complete row provenance in H-MEM.

Public source: github.com/AndiKrue/fasttrack-termin